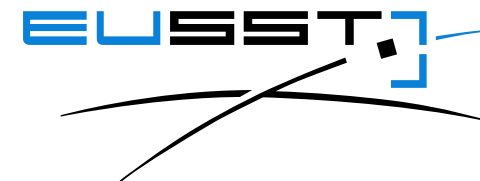


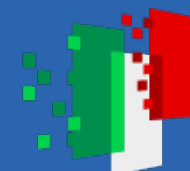
MAJOR UPGRADES TO ITALIAN RADIOTELESCOPES (...and SETI SYSTEMS)



Finanziato
dall'Unione europea
NextGenerationEU



Ministero
dell'Università
e della Ricerca



Italiadomani
PIANO NAZIONALE
DI RIPRESA E RESILIENZA

Northern Cross

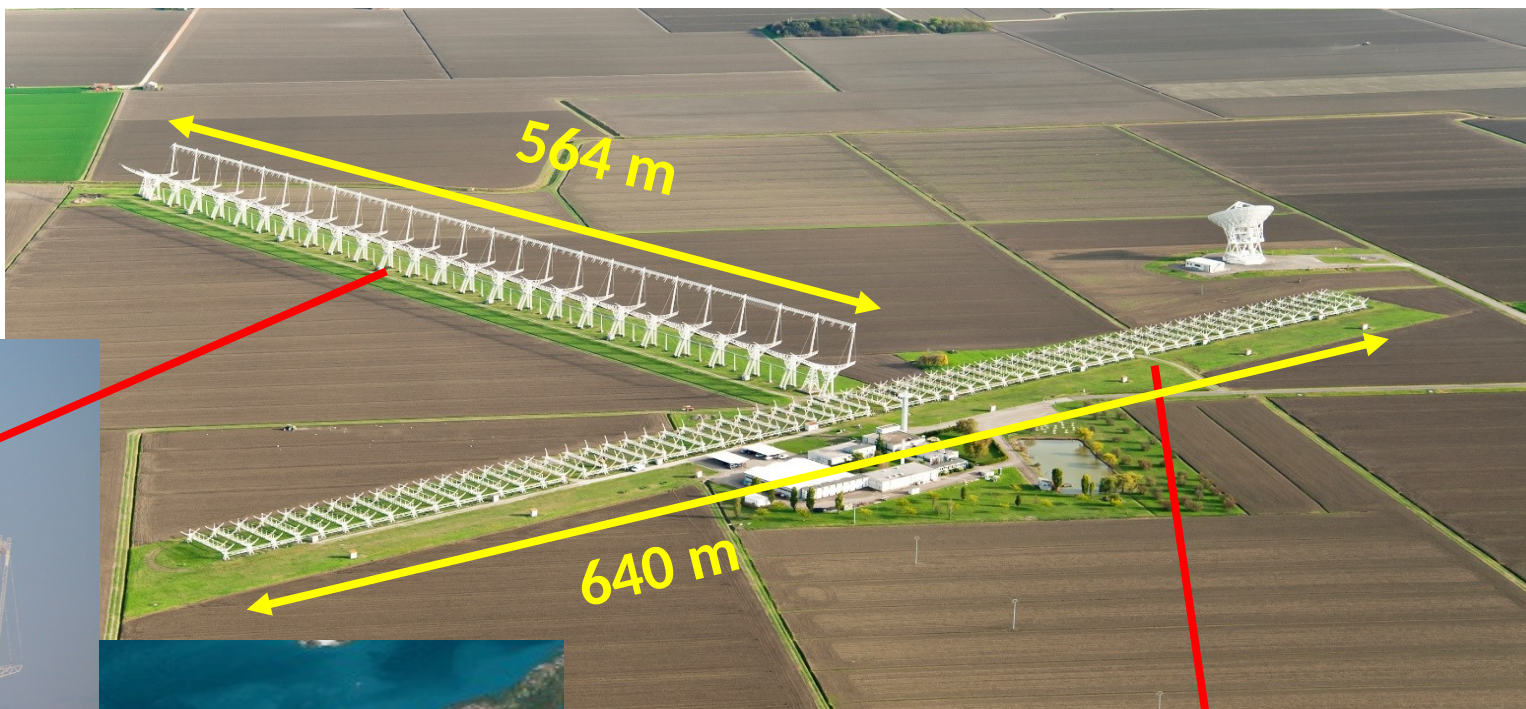
East-West arm

Collecting area = 20.000 sqm

FoV = $26^\circ \times 2^\circ$ (EW x NS)

Resolution = 4 arcmin

Operating frequency = 400-416 MHz



North-South arm

Collecting area = 10.000 sqm

FoV = $7^\circ \times 45^\circ$ (EW x NS)

Resolution = 4 arcmin

Operating frequency = 400-416 MHz



32m «Gavrill Grueff» Radio Telescope



Recent Upgrades (2025)

- «Active Surface System» on primary mirror:
 - 268 mechanical actuators (remotly controlled).
 - 240 new panels
 - Single panels in primary reflector with RMS = 63μm (measured).
 - Total primary reflector RMS ≤ 300μm (to be verified @ high frequacies)
- New «monolithic» Subreflector with RMS = 50μm (measured).
- DBBC3 (Digital Backend for Radio Astronomy)

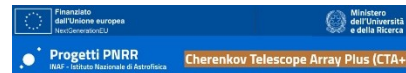
Ongoing Upgrades

- New servosystems (2026)
- 20GHz Optical Link to the control room (2026 - 2027)
- New IFs routing system (2026 - 2027) – partially funded with PNRR - CTA+
- Additional board for DBBC3 (2026-2027)



Ongoing Receivers on Cassegrain (Secondary) Focus:

- CTR – Compact TriBand Receiver (2025)
 - K Band (18-26.5GHz – criogenically cooled)
 - Q Band (33-50GHz – criogenically cooled)
 - W Band(80-116GHz – criogenically cooled)
 - $[W_{\text{LOW}} (80-98\text{GHz}) + W_{\text{HIGH}} (98-116\text{GHz})]$
- CX Band - Dual Feed(2026)
 - 4 - GHz (cryogenically cooled)
- Ku Band - Dual Feed(2026)
 - 13-18GHz (cryogenically cooled)



Installed Receivers on Primary Focus:

- L Band
 - 1.35 - 1.7GHz
- S Band and X Band - Coaxials Receiver
 - 2.2-2.36GHz (cryogenically cooled)
 - 8.18-8.98GHz (cryogenically cooled)

Installed Receivers on Cassegrain (Secondary) Focus:

- C Low Band
 - 4.3-5.8GHz (cryogenically cooled)
- C High Band
 - 5.9-7.1GHz (cryogenically cooled)
- K Band - Dual Feed
 - 18-26.5GHz (cryogenically cooled)

Noto Radio Telescope



Recent Developments

- New simultaneous multi-frequency cryogenic receiver (K-,Q-W- band). ^(a)
- DBBC3 for Interferometric observations. ^(a)
- New «monolithic» Subreflector (50µm surface rms) ^(b)
- New Digital backend (based on Skarab boards for single dish observations) ^(b)
- Uplink to GARR backbone up to 100Gbit/sec

Ongoing Upgrades

- New primary focus (P-band) receiver. Ready in 2026. ^(b)
- Refurbishment of the power distribution and aircolling system of the station. Ready spring 2026. ^(b)
- Refurbishment of the mechanical components of the radiotelescope (included azimuth rail and elevation track). Ready sprint 2026. ^(b)
- New C-,X- band - Dual Feed , cryogenic receiver. Ready 2026. ^(b)



Radiotelescope operative again May/June 2026



(B)



Installed Receivers on Primary Focus:

- L Band
- S Band and X Band - Coaxials Receiver

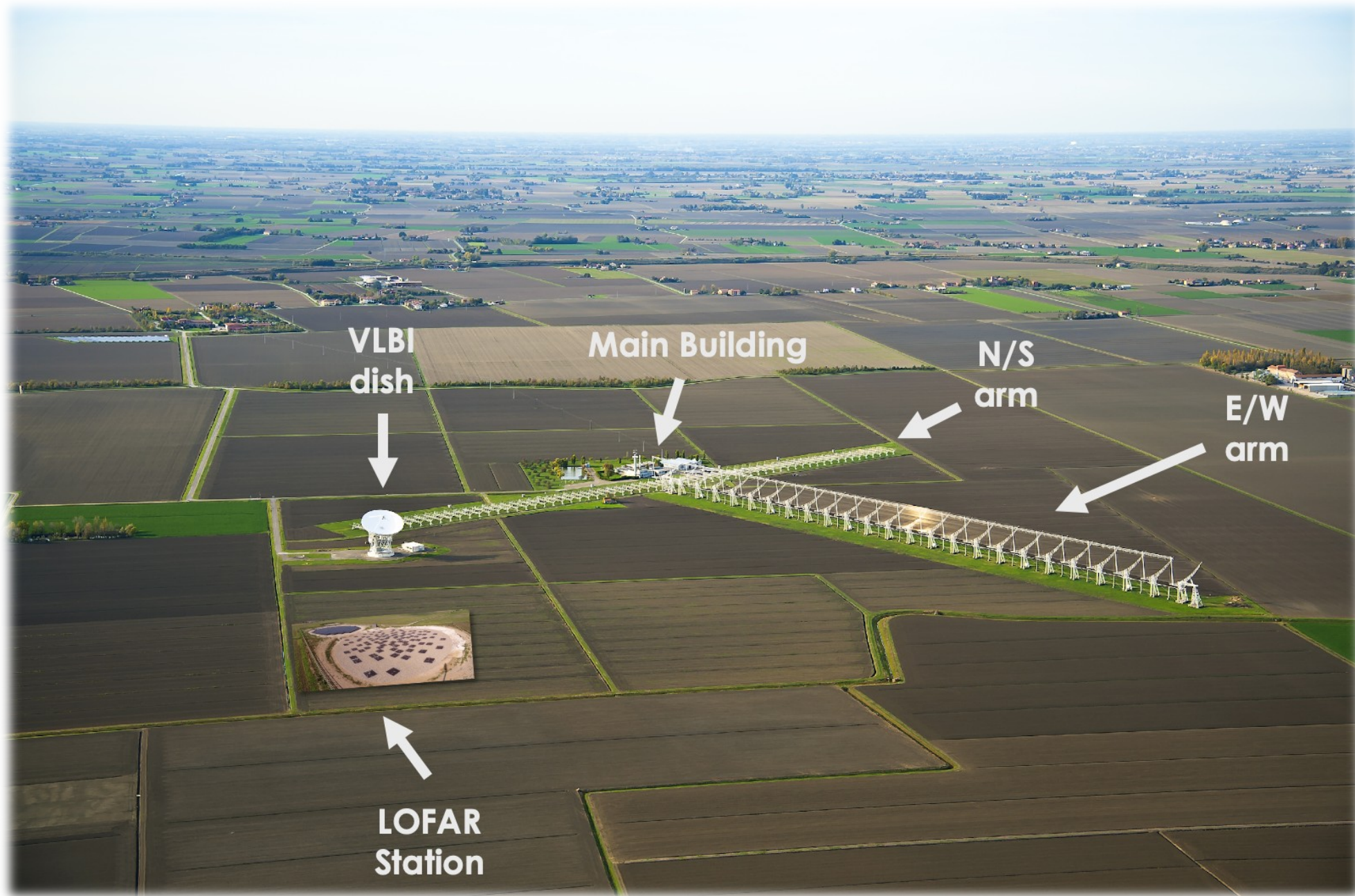
1.35 - 1.7GHz
2.2-2.36GHz (cryogenically cooled)
8.18-8.98GHz (cryogenically cooled)

Installed Receivers on Cassegrain (Secondary) Focus:

- C Low Band
- C High Band
- K Band - Dual Feed

4.3-5.8GHz (cryogenically cooled)
5.9-7.1GHz (cryogenically cooled)
18-26.5GHz (cryogenically cooled)

New International LOFAR2 station



Medicina SETI facilities

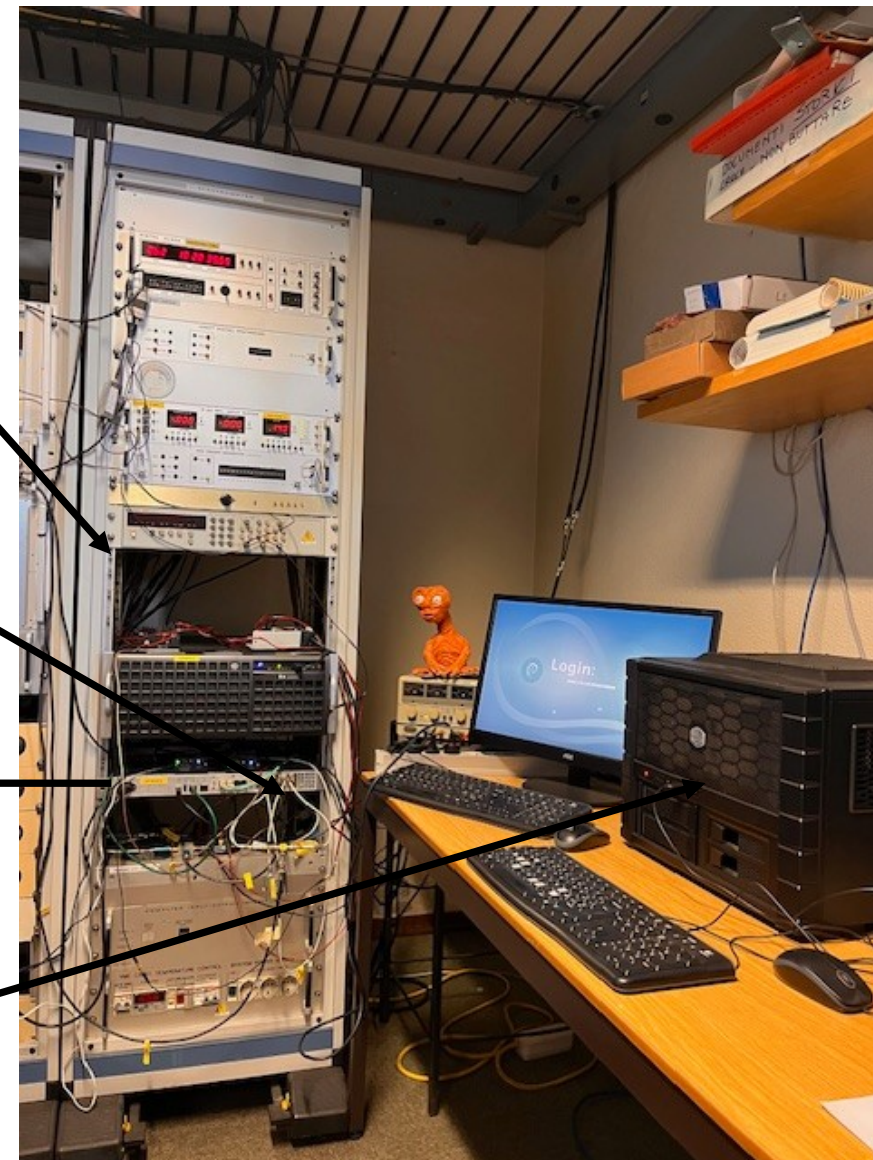


2xRF polarization signal
BW: 500Mhz

Serendip VI: spectrum analyzer:
512M Channel
~1 Hz resolution
NVIDIA TITAN Xp/PCIe/SSE2 x 2

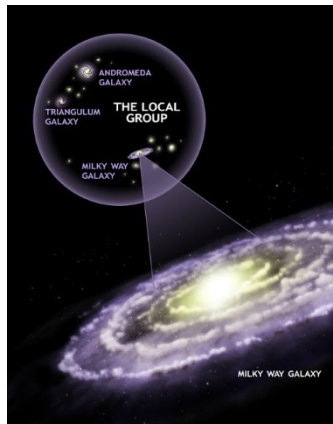
CASPER Roach2:
Virtex-6 SX475T FPGA
PowerPC 440EPx

Nebula Pc:
Intel® Core™ i7-3820 CPU @ 3.60Ghz x 4
NVIDIA TITAN Xp/PCIe/SSE2 x 2



Target	R.A.	Decl.	Year	Distance (kly)
UMa II	08h 51m 30.0s 8.858333	+63° 07' 48 63.13	J2000	98
Leo V	11h 31m 09.60s 11.519333	+02° 13' 12.0s 2.22	J2000	570
Galassia Nana dei Cani da Caccia	13h 28m 03.5s 13.467639	33° 33' 21" 33.555833	J2000	720
Galassia Nana dell'Orsa Minore	15h 09m 08,5s 15.152361	+67° 13' 21" 67.2225	J2000	200
Galassia Nana del Drago	17h 20m 12,4s 17.336778	+57° 54' 55" 57.915278	J2000	260
SGR 1900+14	19h 07m 14.0s 19.120556	+09° 19' 18" 9.321667	J2000	20
Galassia Nana dei Pesci II	22h 58m 31s" 22.975278	+05° 57' 09" 5.9525	J2000	585
Triangulum (M33)	01h 33m 50.89s 1.564136	+30° 39' 36.63" +30.660175	J2000	2900

Target	R. A.	Decl.	Year	Distance (kly)
Segue 1	10h 07m 03.2s 10.117556	+12° 47' 30" +12.791667	J2000	75
UMa III	11h 38m 49.8s 11.647167	+31° 04' 42" +31.078333	J2000	32
COMA (ACO 1656)	12h 59m 44.40s 12.995667	+27° 58' 44.9" +27.979139	J2000	336.000
DRACO II	15h 52m 47s.59 15.879886	+64°33'55" +64.565278	J 2000	?
Galassia Nana del Drago	17h 20m 12,4s 17.336778	+57° 54' 55" +57.915278	J 2000	260.000
Galassia nana di Cassiopea	23h 26m 31.7s 23.442139	+50° 40' 33" +50.675833	J2000	2580
Andromeda Galaxy	00h 42m 44.330 0.712314	+41° 16' 07.50" 41.26875	J2000	2500
Segue 2	02h 19m 06s 2.318333	+20° 10' 31" 20.175278	J2000	114



List of observed galaxies distributed over a 24-hour period

Textbooks in
Science and
Mathematics

TELECOMMUNICATIONS KLT AND RELATIVITY

Volume 1

Claudio Maccone

IPI Press
Library Softbound Edition

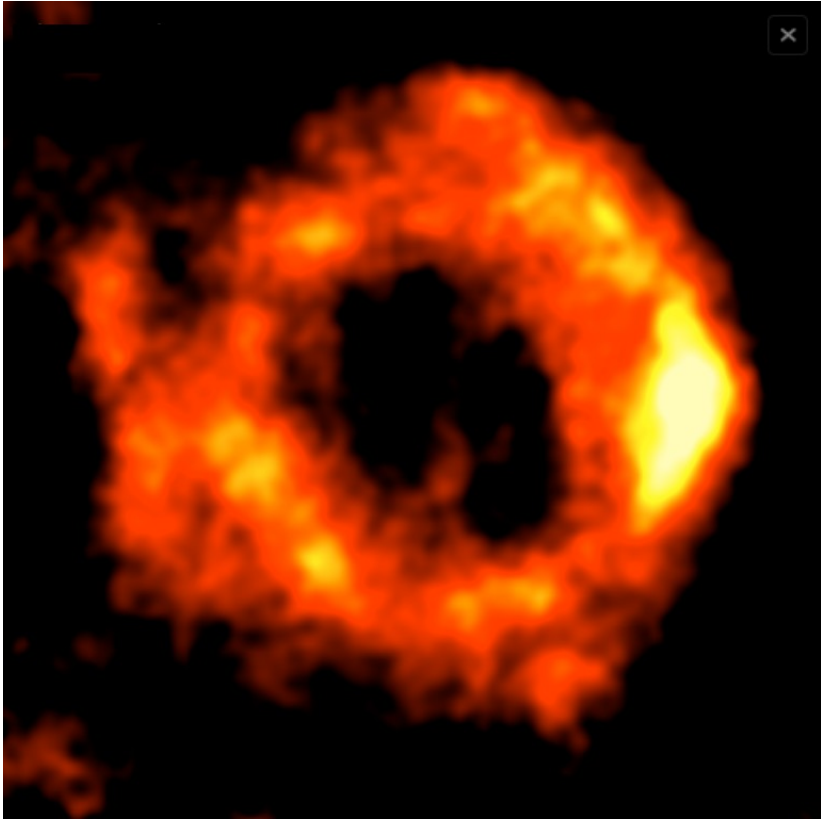
It's probably time for a paradigm shift.

Until now, we have considered searching for narrow-band or limited-band signals as a composition of Sines and Cosines. From the presentations we have seen (Vishal et al.), but also considering our evolution towards digital terrestrial transmissions that are typically broadband, perhaps it's time to abandon the old FFT.

Claudio Maccone had intuited this when he proposed KLT as a new detection algorithm, so probably a new effort to design new-generation machines that use more advanced algorithms could offer more possibilities in the detection of artificial signals of extraterrestrial origin.



- The Sardinia Radio Telescope is under maintenance. We are planning to get new data in 2026, focusing on high-frequencies (Q-band, W-band).



Cassiopeia A imaged at 90 GHz with the MISTRAL receiver at the Sardinia Radio Telescope

- SETI Italy Workshop 2027, scheduled just before the IAC 2027 in Poznan (Poland).